

Are you familiar with Jupyter Notebooks?

Jupyter Notebooks are an interactive tool that combines executable code, explanatory text, visualizations, and other elements into a single document. Widely used in data science, machine learning, and computational analysis, they support multiple programming languages, with Python being the most popular. Their intuitive interface simplifies data exploration, experiments, and real-time documentation.

Here, we have text cells and code cells serve distinct purposes for organizing and presenting content within notebooks:

Text Cells

- These are used for adding explanations, descriptions, and formatting using Markdown or HTML.
- You can insert headings, lists, links, equations, and other elements to document your work and make it more comprehensible (for you and other people who want to use them).

Add a Text Cell:

1. Click the + Text button in the top toolbar.
2. Enter your text in the cell using Markdown for formatting (e.g., # for headings, ** for bold text, * for italics).
3. Click outside the cell or press Shift + Enter to render the formatted text.

Jupyter Cells

Code Cells

- These are used to write and execute programming code, primarily in Python.
- They allow you to test algorithms, manipulate data, and create visualizations, with outputs displayed directly below the executed code.

Add a Code Cell:

1. click the + Code button at the top toolbar of the notebook to insert a code cell below the active one.
2. Enter your code in the cell and press *Shift + Enter* to execute it.

```
In [ ]: #test python code here
test = 4
```

Here I can write **beautiful** texts

Notes:

1: If you want to see the webpages/videos inside this notebook, you need to add this extension:

[Google extension](#) or [Firefox Extension](#)

2: If you want to create a Colab notebook with an R kernel, you can do it with this link:

[Colab with R](#) or

[Other form](#)

Google Colaboratory

Google Colab is a free cloud-based platform that allows you to create, run, and share Jupyter notebooks directly in your browser. It supports languages like Python and provides access to powerful computational resources such as GPUs and TPUs, making it ideal for machine learning and data science tasks.

Additionally, it integrates with Google Drive, enabling easy storage and real-time collaboration.

```
In [ ]: %%html
<iframe width="1000" height="606" src="https://www.youtube.com/embed/inN8seMm7UI
```

Exploring Single-Cell RNA-seq repositories

In this activity, we will explore online repositories and tools for single-cell RNA-seq data analysis. We will navigate through various databases, including the [Single Cell Expression Atlas](#), [Human Cell Atlas Data Portal](#), [CELLXGENE](#), [SRA](#), [GEO](#), [PanglaoDB](#), [CellType](#) and [CellTypist](#), to discover and explore single-cell RNA-seq datasets. Through this hands-on exercise, you will learn how to access, visualize, and interpret single-cell RNA-seq data using online resources.

Objectives:

- Explore single-cell RNA-seq data repositories and tools
- Learn how to access and visualize single-cell RNA-seq data
- Understand how to interpret single-cell RNA-seq data

Note: This activity is designed to be completed in a self-paced manner, and you can work through the exercises at your own speed.

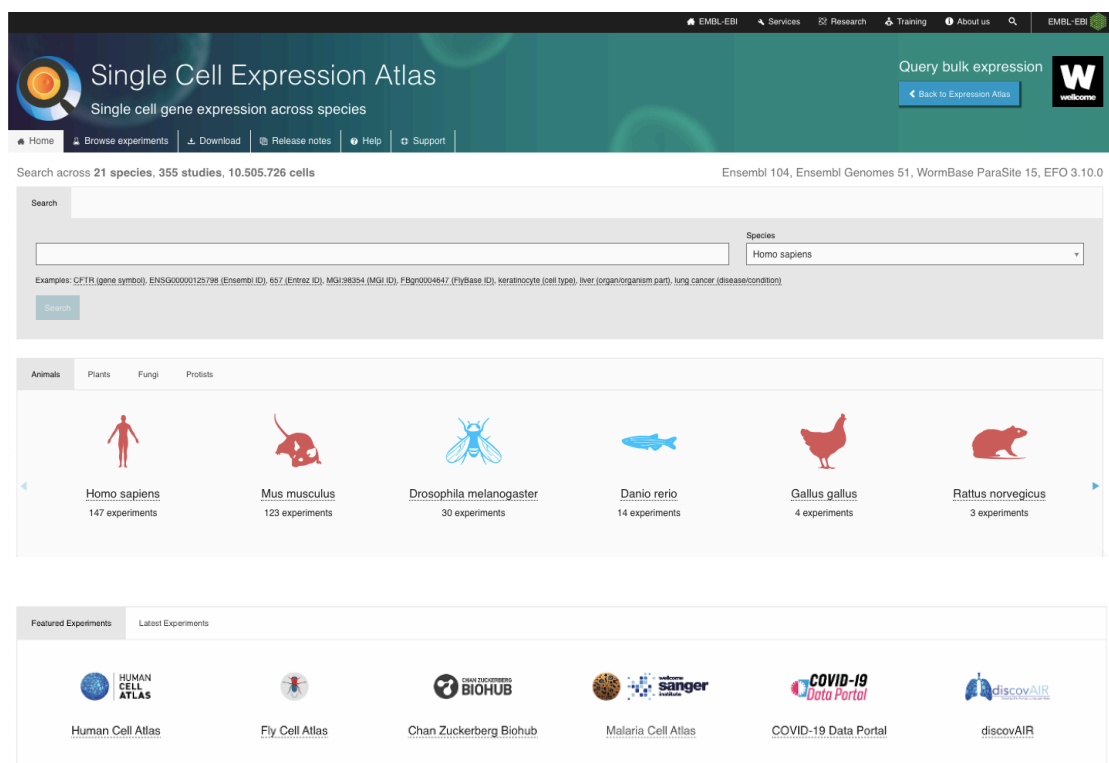
Single Cell Expression Atlas

The Single Cell Expression Atlas is a web-based repository that provides access to a vast collection of single-cell RNA-seq datasets from various organisms and tissues. The atlas allows users to explore and compare gene expression profiles across different cell types, tissues, and conditions.

Practical Exercises:

1. Explore the Single Cell Expression Atlas General Interface:

- Go to the Single Cell Expression Atlas website. You can access the website through your own browser or the browser embedded in the end of this session in this notebook: <https://www.ebi.ac.uk/gxa/sc/home>
- One of the advantages of this repository is the wide range of organisms with single-cell data available, covering Animals, Plants, Fungi and Protists species. Navigate through the webpage, exploring the different species available and explore the one of your interest to further explore.
- Explore also the other set of experiments available, which includes data from the Human Cell Atlas, Fly Cell Atlas, Malaria Cell Atlas, COVID-19 Data Portal and discovAIR initiatives.



2. Search for a Dataset of Interest:

- By selecting an species or experiment collection of interest, you will have access to a list of experiments linked to that organism/initiative.
- There you can select experiments based on different variables, including the organism Kingdom, Experiment Collection, Technology Type. You can also explore other features such as the Title of the study, Experimental Factors (e.g.

organism part, developmental state, age, etc), and the Number of Cells available.

- There you can also select the set of experiments of interest to download the Raw quantification files, Normalised counts files and Experiment design file.

The screenshot shows the Single Cell Expression Atlas website. The header includes the logo, title "Single Cell Expression Atlas", subtitle "Single cell gene expression across species", and a "Query bulk expression" button with a "Back to Expression Atlas" link and the Wellcome logo. The navigation bar has links for Home, Browse experiments, Download, Release notes, Help, and Support. Below the navigation bar are filters for Kingdom (All), Experiment collection (All), Technology type (All), and Entries per page (10). A search bar is also present. The main content area displays a table of experiments with columns for Load date, Title, Experimental factors, Number of cells, and Download 3 entries. The table lists several experiments, including "Comparative analysis of kidney organoid and adult human kidney single cell and single nucleus transcriptomes - human kidney subset", "Single cell transcriptome atlases of the developing mouse and human spinal cord - human data", "scRNA-seq analysis of human mammary epithelial cells derived from non-lactating breast tissue or from milk", "Single-cell RNA-seq of human peripheral blood mononuclear cells (PBMCs) throughout the febrile, critical, and convalescence phases of dengue virus infection", "Single cell RNA-seq of in vitro cultured human embryos", "The atlas of the human hypertrophied heart reveals impaired cell communication", and "Single-Cell RNA Sequencing of Retinal Organoids Derived by CSC14 hESCs". A modal window is open over the table, showing options to download "Raw quantification files", "Normalised counts files", and "Experiment design file". The modal has "Cancel" and "Download" buttons.

Load date	Title	Experimental factors	Number of cells	Download 3 entries
16-01-2023	Comparative analysis of kidney organoid and adult human kidney single cell and single nucleus transcriptomes - human kidney subset	• organism part	1982	☒
14-09-2022	Single cell transcriptome atlases of the developing mouse and human spinal cord - human data	• sampling site • developmental stage	117640	☒
19-08-2022	scRNA-seq analysis of human mammary epithelial cells derived from non-lactating breast tissue or from milk	• phenotype • organism part	58870	☒
19-08-2022	Single-cell RNA-seq of human peripheral blood mononuclear cells (PBMCs) throughout the febrile, critical, and convalescence phases of dengue virus infection	• clinical history • sampling time point • inferred cell type - authors labels • inferred cell type - ontology labels	39638	☐
08-08-2022	Single cell RNA-seq of in vitro cultured human embryos	• age	138041	☐
07-08-2022	The atlas of the human hypertrophied heart reveals impaired cell communication	• individual	19053	☐
07-08-2022	Single-Cell RNA Sequencing of Retinal Organoids Derived by CSC14 hESCs	• time	20861	☐

3. Visualize and Explore Available Data:

- Select a gene or cell type of interest and navigate through the interactive browser, exploring different aspects of the data available.
- Select datasets based on the "Inferred cell type - ontology labels" or "Organism part".
- Select a dataset of interest and dive into all the data available.
- Visualize the gene expression profiles using the interactive plots (e.g., t-SNE, heatmap).
- Visualize gene markers and enjoy other nice features from the repository.

Gene ID or gene symbol
CFTR

Species
Any

Search

Species
☐ Danio rerio
☐ Homo sapiens
☐ Mus musculus

Inferred cell type - ontology labels
Select...

Organism part
Select...

CFTR is expressed in:

Species	Marker genes	Title	Experimental variables	Number of assays
Mus musculus	- See cluster 15 for k = 80 - See cluster 19 for k = 50 - See cluster 20 for k = 67 - See cluster 10 for k = 21 - See cluster 15 for k = 37 - See cluster 13 for k = 25 - See cluster 14 for k = 69 - See cluster 15 for k = 29	Single-cell RNA-seq of nasal cavity mucosa cells of mice stimulated with interferon alpha	- stimulus - inferred cell type - authors labels - inferred cell type - ontology labels	
Homo sapiens	- See cluster 11 for k = 32 - See cluster 11, 15 for k = 37 - See cluster 5, 9 for k = 58 - See cluster 15, 7 for k = 74 - See cluster 11, 13, 9 for k = 46 - See cluster 11, 6 for k = 37	Single-cell RNA sequencing of the human biliary tree	- organism part - growth condition - stimulus	

Home Browse experiments Download Release notes Help Support

Single-cell RNA sequencing of the human biliary tree

Single-cell RNA-Seq mRNA baseline

Number of cells: 72,803
Organism: Homo sapiens
Publication:
Sampaziotis F, Muraro D, Tysoe OC, Sawiak S, Beach TE et al. (2021) *Cholangiocyte organoids can repair bile ducts after transplantation in the human liver*.

Results Supplementary information Downloads

Cell plots

Marker Genes

Anatomogram

Gene information

Plot type: UMAP

Plot options: n_neighbors: 30

Colour plot by: k = 37

Gene ID, gene name or gene features: ENSG00000011026

Clusters

Gene expression

Cluster 1 Cluster 2 Cluster 3 Cluster 4 Cluster 5 Cluster 6 Cluster 7 Cluster 8 Cluster 9 Cluster 10 Cluster 11 Cluster 12 Cluster 13 Cluster 14 Cluster 15 Cluster 16 Cluster 17 Cluster 18 Cluster 19 Cluster 20 Cluster 21 Cluster 22 Cluster 23 Cluster 24 Cluster 25 Cluster 26 Cluster 27 Cluster 28 Cluster 29 Cluster 30 Cluster 31 Cluster 32 Cluster 33

Expression level (CPM)

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Single-cell RNA sequencing of the human biliary tree

Single-cell RNA-Seq mRNA baseline

Number of cells: 72,803
Organism: Homo sapiens
Publication:
Sampaziotis F, Muraro D, Tysoe OC, Sawiak S, Beach TE et al. (2021) *Cholangiocyte organoids can repair bile ducts after transplantation in the human liver*.

Results Supplementary information Downloads

Cell plots

Marker Genes

Anatomogram

Gene information

Number of clusters: k = 37

Show marker genes for: All clusters

Cluster marker genes

Download

Median expression (CPM)

0.1 1 10 100 1k 10k 100k

Cluster 1 Cluster 2 Cluster 3 Cluster 4 Cluster 5 Cluster 6 Cluster 7 Cluster 8 Cluster 9 Cluster 10 Cluster 11 Cluster 12 Cluster 13 Cluster 14 Cluster 15 Cluster 16 Cluster 17 Cluster 18 Cluster 19 Cluster 20 Cluster 21 Cluster 22 Cluster 23 Cluster 24 Cluster 25 Cluster 26 Cluster 27 Cluster 28 Cluster 29 Cluster 30 Cluster 31 Cluster 32 Cluster 33

RP29P
ENSG00000211814
RPL309A
RPL7
PLIN2
RPL2
PTAB1
RPL1
RPL24
ENSG00000260836
RPL12
RPL23
TNXB10
CTSL
RPLP0
TNXB1
RPL2A
CC120
ENSG00000242343
ENSG00000230382
RPL1

Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5

[Home](#)
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Search term

keratinocyte

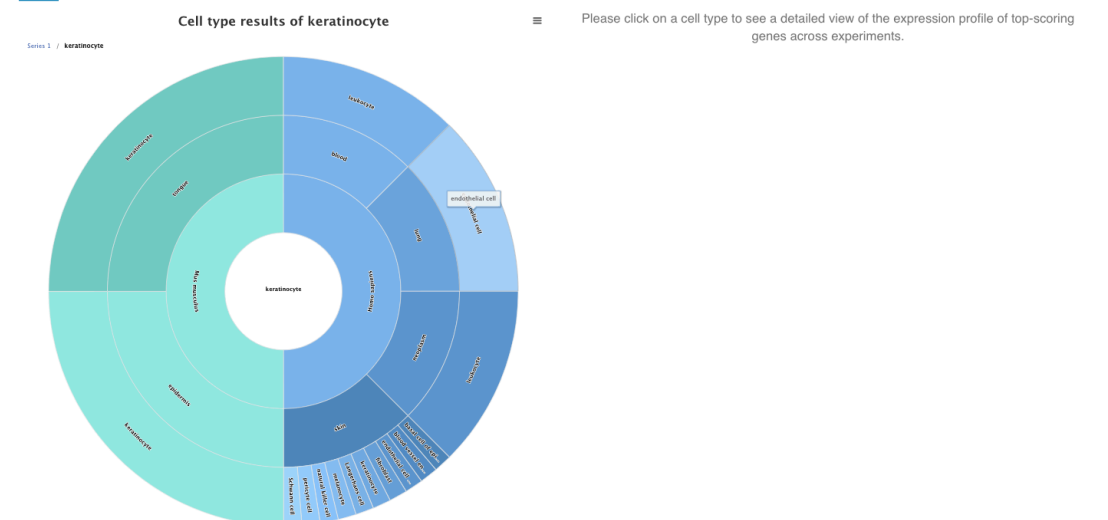
×

Species

Any

▼

Search



4. Now enjoy the repository, by playing with the data and genes of interest.

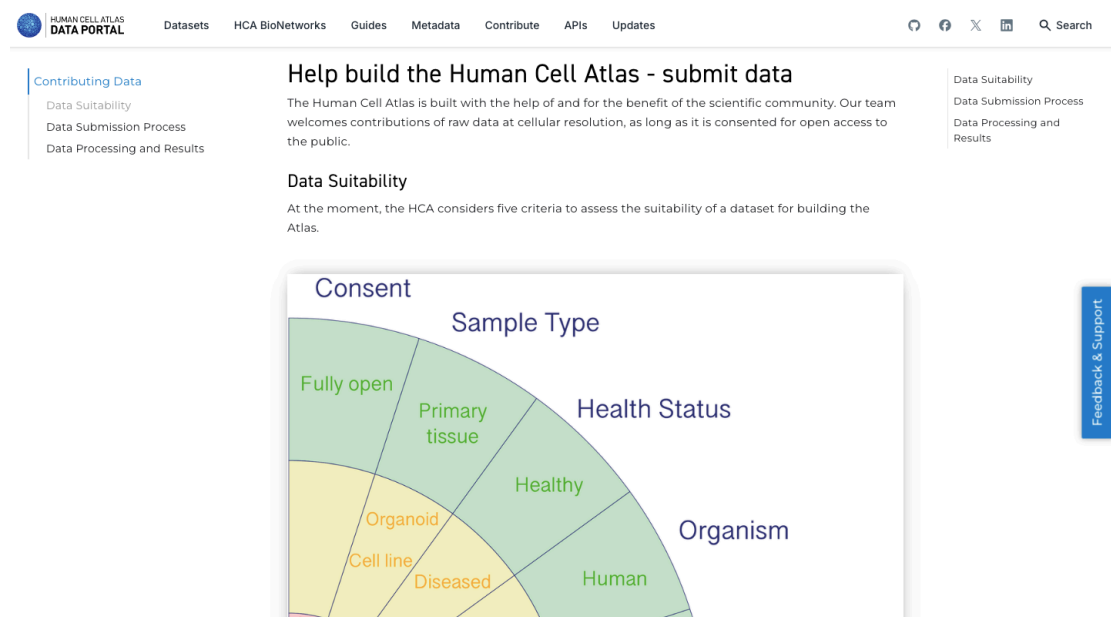
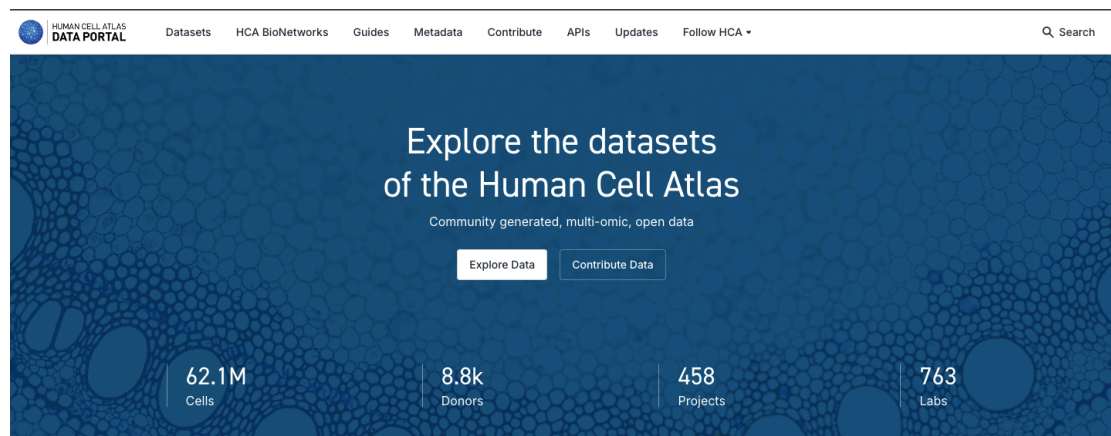
Brief Description:

The Human Cell Atlas Data Portal is a web-based repository that provides access to a vast collection of single-cell RNA-seq datasets from various human tissues and cell types. The portal is the main repository of the Human Cell Atlas initiative, and it allows users to explore, visualize, and analyze gene expression profiles across different cell types, tissues, and conditions generated by the consortium.

Practical Exercises:

1. Explore the General Portal Interface:

- Go to the Human Cell Atlas Data Portal website:
<https://data.humancellatlas.org/>
- Explore the website general functionalities, visualizing the data available and how to contribute with data.



2. Search and Explore a Dataset of Interest:

- Explore the datasets available, navigating through the Projects, Samples and Files available. Use the filters available to perform a more specific selection.

- After selecting a particular dataset, navigate through the dataset Overview, Metadata, Matrices, Download and Export tabs.
- Check if the dataset allow you to explore it deeper in another Analysis Portal (e.g. UCSC Genome Browser or CELLXGENE).
- Check the possibility to Export the data available to analyze it in Terra cloud solution. Note that this is a private third party platform.

The screenshot shows the 'Human Cell Atlas Data Explorer' interface. On the left is a 'Filters' sidebar with categories like Project, Donor, and Sample. The main area is titled 'Explore Data' and shows a table of results. At the top right, it displays summary statistics: 62.1M Estimated Cells, 21.9k Specimens, 8.8k Donors, and 520.4k Files. An 'Export' button is visible.

Project Title	Species	Anatomical Entity	Library Construction Method	Disease (Donor)	Cell Count Estimate	Updated
Single-cell RNA-seq reveals ectopic and aberrant lung-resident cell populations in idiopathic pulmonary fibrosis	Homo sapiens	lung	10x 3' v2	3 diseases	312.928	2024-07-0
Dynamic scRNA-seq of live human pancreatic slices reveals functional endocrine cell neogenesis through an intermediate ducto-acinar stage.	Homo sapiens	pancreas	10x 3' v2	normal	53.950	2024-06-2
Dandelion uses the single-cell adaptive immune receptor repertoire to explore lymphocyte developmental origins	Homo sapiens	blood	10x immune profiling	normal	17.887	2024-06-2
Genetic ancestry effects on the response to viral infection are pervasive but cell type specific	Homo sapiens	blood	10x 3' v2	normal	235.000	2024-06-2

The screenshot shows the 'Overview' tab for a dataset titled 'Single-nucleus cross-tissue molecular reference maps toward understanding disease gene function.' It includes a description, contact information, and analysis portals.

Description: Understanding gene function and regulation in homeostasis and disease requires knowledge of the cellular and tissue contexts in which genes are expressed. Here, we applied four single-nucleus RNA sequencing methods to eight diverse, archived, frozen tissue types from 16 donors and 25 samples, generating a cross-tissue atlas of 209,126 nuclei profiles, which we integrated across tissues, donors, and laboratory methods with a conditional variational autoencoder. Using the resulting cross-tissue atlas, we highlight shared and tissue-specific features of tissue-resident cell populations; identify cell types that might contribute to neuromuscular, metabolic, and immune components of monogenic diseases and the biological processes involved in their pathology; and determine cell types and gene modules that might underlie disease mechanisms for complex traits analyzed by genome-wide association studies.

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 Broad Institute of MIT and Harvard
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 Orit Rozenblatt-Rosen

Analysis Portals:
 CZ CELLxGENE
 UCSC Cell Browser

Project Details:
 Project Label: CrossTissueReferenceMap
 Species: Homo sapiens
 Sample Type: specimens
 Anatomical Entity: 7 anatomical entities
 Organ Part: 6 organ parts
 Selected Cell Types

3. Search and Explore the Atlases Generated by a Human Cell Atlas Bionetwork:

- Note that each Bionetwork possesses datasets, but at the moment only Lung and Nervous System have unified atlases available.
- Navigate through the different bionetworks, exploring the data and atlases available.
- Explore the different particularities of the components of the atlases, such as the number of tissues, disease state, number of cells and the possibility to

explore it further in CELLxGENE or downloading the data.

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Search

HCA Biological Network Atlases

Adipose

Gut

Lung

Pancreas

Breast

Heart

Musculoskeletal

Reproduction

Development

Immune

Nervous System

Skin

Eye

Kidney

Oral and Craniofacial

Genetic Diversity

Liver

Organoid

[Datasets](#)[HCA BioNetworks](#)[Guides](#)[Metadata](#)[Contribute](#)[APIs](#)[Updates](#)

HCA Biological Network Atlases > Gut Network

Gut Network

[Network Overview](#)[HCA Gut Datasets](#)

Network Description

The intestine is one of the most complex organs in the human body and serves a wide range of functions, including the digestion and absorption of nutrients and a major site of immune interactions. Different anatomical sections of the gut play specific roles in the digestive process, which are performed by different tissue compositions or cell types in each section that communicate with each other in myriad ways. Interactions with trillions of nearby microbes add further complexity.

The Gut Biological Network of the Human Cell Atlas aims to generate a detailed map of the cells forming the intestinal tube, thereby providing critical information that will allow us to further unravel the complex networks that operate in health. Importantly, understanding the healthy gut provides the starting point for interrogating related diseases such as bowel cancer and inflammatory bowel diseases, which affect an increasing number of patients.

An initial roadmap for the Gut Cell Atlas was released in 2021 with funding from the Helmsley Charitable Trust. The roadmap was subsequently updated and expanded by the HCA Gut Biological Network and will be released in Spring 2023.

0
Atlases

0
Diseases

0
Tissues

0
Cells

Network Coordinators

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Jose Ordovas-Montanes
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Gut Network atlases are still under development.

Meanwhile, individual [HCA Gut Datasets](#) are available as potential source datasets for integrated atlases.

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HCA Biological Network Atlases > Gut Network

Gut Network

[Network Overview](#)[HCA Gut Datasets](#)

The HCA datasets below focus on the network's tissues and are potential source datasets for integrated atlases.

Project Title	Species	Method	Anatomical Entity	Disease (Donor)	Cell Count Est.
A cell atlas of human thymic development defines T cell repertoire formation	2 species	4 library cons...	2 anatomical ...	normal	250,000
A human cell atlas of fetal gene expression.	Homo sapiens	sci-RNA-seq3	14 anatomical...	normal	4,062,965
A proximal-to-distal survey of healthy adult human small intestine and colon epithelium by single-cell transcriptomics	Homo sapiens	10x 3' v3	2 anatomical ...	normal	12,590
A single-cell atlas of chromatin accessibility in the human genome	Homo sapiens	sci-RNA-seq	18 anatomical...	normal	500,000
An organoid and multi-organ developmental cell atlas reveals multilineage fate specification in the human intestine	Homo sapiens	2 library cons...	5 anatomical ...	Unspecified	Unspecified

 Lung Network

Network Overview

Network Description

The lung is the organ essential for gas exchange, which is facilitated by a highly specialized anatomy involving distinct epithelial and endothelial cell populations, supported by various mesenchymal cell types providing structural support as well as a niche guiding epithelial differentiation. In addition, the lung is an essential part of the mucosal immune system, harboring a rich variety of specialized tissue-resident immune cells.

The high cellular complexity of lung tissue has resulted in the description of at least 60 different cell types to date. Regional differences in the molecular states of these cells as these occur along the various gradients and axes of the lung further adds to the cellular heterogeneity in this intricate tissue.

The Lung Biological Network of the HCA aims to establish a complete atlas of the healthy human lung. This atlas will sample the entire coordinate framework of the lung, both macro- and micro-anatomically, to capture the natural variation of cellular states present within the healthy population, as outlined in this paper. This effort will generate a comprehensive map that describes the lung cells and their molecular phenotypes, the transitions between different cell states and their location in the tissue relative to each other and to anatomical landmarks.

The Human Lung Cell Atlas will serve as a reference for the analysis of diseased lung tissue at single-cell resolution and will allow identification of the shifts in cellular repertoire, the changes in cellular states and phenotypes, and the altered cell-cell interactions that disrupt normal lung homeostasis and constitute disease.

1 Atlases	15 Diseases
4 Tissues	2.4M Cells

Network Coordinators

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Martijn Nawijn
Jay Rajagopal
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lung@humancellatlas.org

Atlases

Atlas Name	Tissue	Disease	Assay	Cells
The Integrated Human Lung Cell Atlas (HLCA) v1.0	4 tissues	normal 15 diseases	9 assays	2.4M

The integrated Human Lung Cell Atlas (HLCA) v1.0

Atlas Overview

[Source Datasets](#)

Atlas Description

The integrated Human Lung Cell Atlas (HLCA) represents the first large-scale, integrated single-cell reference atlas of the human lung. It consists of over 2 million cells from the respiratory tract of 486 individuals and includes 49 different datasets. It is split into the fully integrated HLCA core reference and the extended or full HLCA.

HLCA Core

The HLCA core includes data of healthy lung tissue from 107 individuals and includes manual cell type annotations based on consensus across six independent experts, as well as demographic, biological, and technical metadata. The datasets in the HLCA core were integrated using scANVI. The HLCA core can be used as a reference to map new datasets onto using scArches, allowing the transfer of the harmonized cell-type labels to any new dataset.

HLCA Full

The full HLCA includes 35 further datasets that include donors with various lung diseases. These datasets were mapped onto the core with scArches, and include disease annotations as well as consensus cell type labels transferred from the HLCA core onto the mapped datasets.

Batch Correction

Note that while the HLCA includes an integrated, batch-corrected low-dimensional embedding, the gene counts themselves were not batch-corrected. Both the HLCA core and the full HLCA can be explored below.

Metadata

Detailed information about all metadata in the objects, as well as further HLCA-related information, can be found on the HLCA landing page: <https://github.com/LungCellAtlas/HLCA>.

Raw Counts

Raw counts are available in the downloaded .h5ad at `adata.raw.X`, in the downloaded .rds at `seurat_object@assays$RNA@counts`, and via CELLxGENE Census.

HCA Biological Network Atlases

 [Lung Network](#)

Publication

[Sikkema et al. \(2023\) Nat Med](#)

Code

<https://github.com/LungCellAtlas/HLCA>

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Downloaded and exported data is governed by the
[HCA Data Release Policy](#) and licensed under the
[Creative Commons Attribution 4.0 International](#)

HUMAN CELL ATLAS DATA PORTAL

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The Integrated Human Lung Cell Atlas (HLCA) V1.0

[Atlas Overview](#)
[Source Datasets](#)

The datasets below are ingested in the HCA Data Portal and used in the creation of the related integrated atlas. Additional datasets may have been used in the creation of the atlas. See the atlas publication for the full list of source datasets for the atlas. Additional lung network datasets can be found in the [HCA Data Explorer](#).

Project Title	Species	Method	Anatomical Entity	Disease (Donor)	Cell Count Est.
A cellular census of human lungs identifies novel cell states in health and in asthma	Homo sapiens	2 library cons...	2 anatomical ...	normal chronic obstructive pulmonary disease	47,291
A molecular cell atlas of the human lung from single-cell RNA sequencing	2 species	2 library cons...	2 anatomical ...	normal 3 diseases	75,000
A Single-Cell Atlas of the Human Healthy Airways	Homo sapiens	10× 3' v2	3 anatomical ...	normal	77,969
A spatially resolved atlas of the human lung characterizes a gland-associated immune niche	Homo sapiens	6 library cons...	lung	normal type 2 diabetes mellitus	193,108
Airspace Macrophages and Monocytes Exist in Transcriptionally Distinct Subsets in Healthy Adults	Homo sapiens	10× 3' v3	lung	normal	49,384

3. Explore the guides available to know more about all functionalities and different aspects of the HCA Data Portal.:

HUMAN CELL ATLAS DATA PORTAL

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[Metadata](#)
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Guides

[Exploring Projects](#)

[Exploring Biological Network and Atlas Data](#)
[Accessing HCA Data and Metadata](#)
[Exporting HCA Data to Terra](#)
[Exploring Project Matrices](#)
[Accessing Metadata via TDR](#)

Exploring Projects

Projects are a basic unit of data organization in the HCA Data Portal. Project contributors contribute raw sequencing and associated [files](#) along with rich metadata describing:

1. the [origin and type of the cells](#) used in the project
2. the [processes and protocols](#) used to collect and process the cells prior to sequencing
3. the [sequencing](#) methods used
4. details about the [project](#) contributors and their institutions

This [Metadata](#) is included in the project's Metadata Manifest (TSV file). When the HCA Data Portal processes the contributor's raw data with uniform pipelines, this processing information is also added to the Metadata Manifest.

Finding a Project of Interest

The Data Portal Explore page lists all projects by title along with key project metadata. The project list is filterable by metadata values.

[Finding a Project of Interest](#)
[Viewing Project Details](#)
[Downloading Project Metadata](#)
[Downloading Project HCA Data Portal-Generated Matrices](#)
[Downloading Project Contributor-Generated Matrices](#)

[Feedback & Support](#)

4. Now enjoy the repository, by playing with the data and genes of interest.

CELLxGENE

Brief Description:

CELLxGENE is a web-based portal developed by the Chan Zuckerberg Initiative (CZI) that enables interactive exploration and analysis of single-cell RNA-seq data. It provides a user-friendly interface to visualize and compare gene expression profiles across different cell types, tissues, and conditions.

Practical Exercises:

1. Explore the General Portal Interface:

- Go to the CELLxGENE website: <https://cellxgene.cziscience.com/>
- CELLxGENE provides a plenty of tools which allows you to further explore and analyze single-cell data.
- Navigate through the Collections and Datasets available and select one of interest to further explore. You can use the "Filters" option to select the data according to different aspects of interest (e.g. cell type, disease, self-reported ethnicity, sex).

APPLICATION: Collections, Datasets, Gene Expression, Cell Guide, Differential Expression **NEW**

CENSUS: API, Models

Help & Documentation

CZ CELLxGENE DISCOVER

Discover the mechanisms of human health

Download and visually explore data to understand the functionality of human tissues at the cellular level with Chan Zuckerberg CELL by GENE Discover (CZ CELLxGENE Discover).

UNIQUE CELLS	DATASETS	CELL TYPES
89.8M	1458	903

Differential Expression Explorer Census Cell Guide Collections & Datasets Gene Expression Annotate

APPLICATION: Collections, Datasets, Gene Expression, Cell Guide, Differential Expression **NEW**

CENSUS: API, Models

Help & Documentation

Filters

- Assay
- Cell Type
- Consortia
- Development Stage
- Disease
- Organism
- Publication
- Publication Date
- Self-Reported Ethnicity
- Sex
- Suspension Type
- Tissue

Collections 225 of 225

Collections	Publication	Tissue	Disease	Organism
iPain atlas	No publication	dorsal root ganglion trigeminal ganglion	normal 2 diseases	Mus musculus
Multimic Profiling of Human Clonal Hematopoiesis Reveals Genotype and Cell-Specific Inflammatory Pathway Activation	Heimlich et al. (2024) Blood Advances	blood	clonal hematopoiesis normal	Homo sapiens
Asian Immune Diversity Atlas (AIDA)	Kock et al. (2024) bioRxiv	blood	normal	Homo sapiens
iPSC-derived small intestine-on-chip with self-organizing epithelial, mesenchymal and neural cells	Moerkens et al. (2024) Cell Reports	intestinal mucosa (organoid)	normal	Homo sapiens
Comparative Analysis of Commercial Single-Cell RNA Sequencing Technologies	De Simone et al. (2024) bioRxiv	blood	normal	Homo sapiens
An Integrated Single Cell Atlas of Human Periodontitis	Easter et al. (2024) Nat Commun	gingiva	normal 2 diseases	Homo sapiens
Single Cell Atlas of Mouse Healthy Gingiva	Easter et al. (2024) Nat Commun	gingiva	normal	Mus musculus
An atlas of healthy and injured cell states and niches in the human kidney (Version 1.5)	No publication	4 tissues	normal 2 diseases	Homo sapiens
HCA kidney seed network: University of Michigan	No publication	cortex of kidney	normal	Homo sapiens
Intratumoral heterogeneity in recurrent pediatric pilocytic astrocytomas	No publication	cerebellum	normal pilocytic astrocytoma	Homo sapiens
Human and mouse dermal fibroblast atlas	No publication	15 tissues	normal	Homo sapiens

APPLICATION

Collections

Datasets

Gene Expression

Cell Guide

Differential Expression

NEW

CENSUS

API

Models

Help & Documentation

Filters

Assay

Cell Count

Cell Type

Consortia

Development Stage

Disease

Gene Count

Organism

Publication

Publication Date

Self-Reported Ethnicity

Sex

Suspension Type

Tissue

Datasets

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Tissue

Disease

Assay

Organism

Cells

Download

Explore

Single Cell Sequencing of Human PBMCs In Clonal Hematopoiesis of Indeterminant Potential

Multitomic Profiling of Human Clonal Hematopoiesis Reveals Genotype and Cell-Specific Inflammatory Pathway Activation

blood

clonal hematopoiesis normal

10x 3' v3

Homo sapiens

66.985

Download

Explore

IPain Atlas DRG Nociceptive Lineage Cells - SCENIC

IPain atlas

dorsal root ganglion

normal peripheral neuropathy, non-human animal

7 assays

Mus musculus

70.963

Download

Explore

IPain Atlas DRG Nociceptive Lineage Cells - TF activity

IPain atlas

dorsal root ganglion

normal peripheral neuropathy, non-human animal

7 assays

Mus musculus

70.963

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Explore

IPain Atlas DRG All Cells - RNA

IPain atlas

dorsal root ganglion

normal peripheral neuropathy

6 assays

Mus musculus

191.798

Download

Explore

IPain Atlas DRG Nociceptive Lineage Cells - RNA

IPain atlas

dorsal root ganglion

normal peripheral neuropathy, non-human animal

7 assays

Mus musculus

70.963

Download

Explore

IPain Atlas TG Nociceptive Lineage Cells - RNA

IPain atlas

trigeminal ganglion

normal peripheral neuropathy

5 assays

Mus musculus

14.057

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Explore

IPain Atlas TG All Cells - RNA

IPain atlas

trigeminal ganglion

normal peripheral neuropathy

5 assays

Mus musculus

84.658

Download

Explore

APPLICATION

Collections

Datasets

Gene Expression

Cell Guide

Differential Expression

NEW

CENSUS

API

Models

Help & Documentation

Asian Immune Diversity Atlas (AIDA)

CZI Single-Cell Biology

Individuals of Asian ancestry constitute 60% of the global population, but have been underrepresented in genomics studies. Given the pervasive influence of dimensions of human diversity such as age, ancestry, and sex on biomedically relevant phenotypes, the construction of healthy, reference atlases for disease comparisons requires equitable and diverse representation from all population groups. In the first phase of the Asian Immune Diversity Atlas, we profiled 1,058,909 peripheral blood mononuclear cells from 503 healthy donors spanning five Asian ancestries across Japan, Singapore, and South Korea alongside common controls via 5'-single-cell RNA sequencing. We elucidated cell type proportion and cell state enrichment signatures and differences across ancestries, ages, and sexes, including interactions between these demographics that influenced single cell phenotypes. We performed single-cell pseudobulk expression quantitative trait loci (eQTL) analyses for 19 immune cell subtypes, identifying more than 11,000 genes associated with such eQTLs, and thereafter integratively analysed these data with datasets from genome-wide association studies to prioritise causal genes and loci and implicated cell subtypes. Our resource is the first healthy reference atlas to include multiple Asian population groups and more broadly to examine the influence of ancestry on variation in single cell phenotypes, with the potential to facilitate precision medicine efforts in Asia.

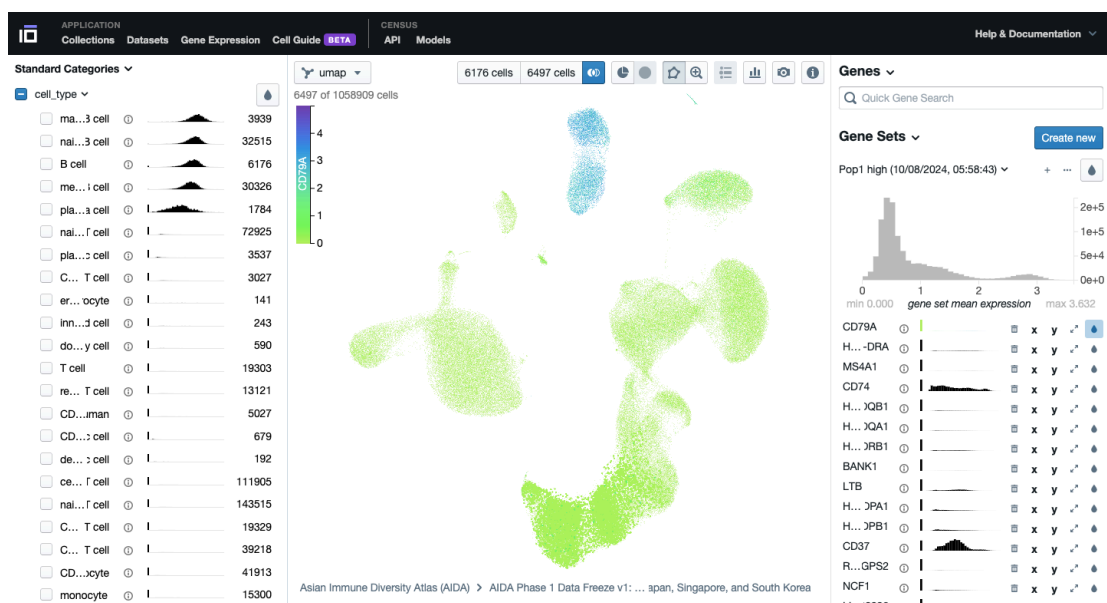
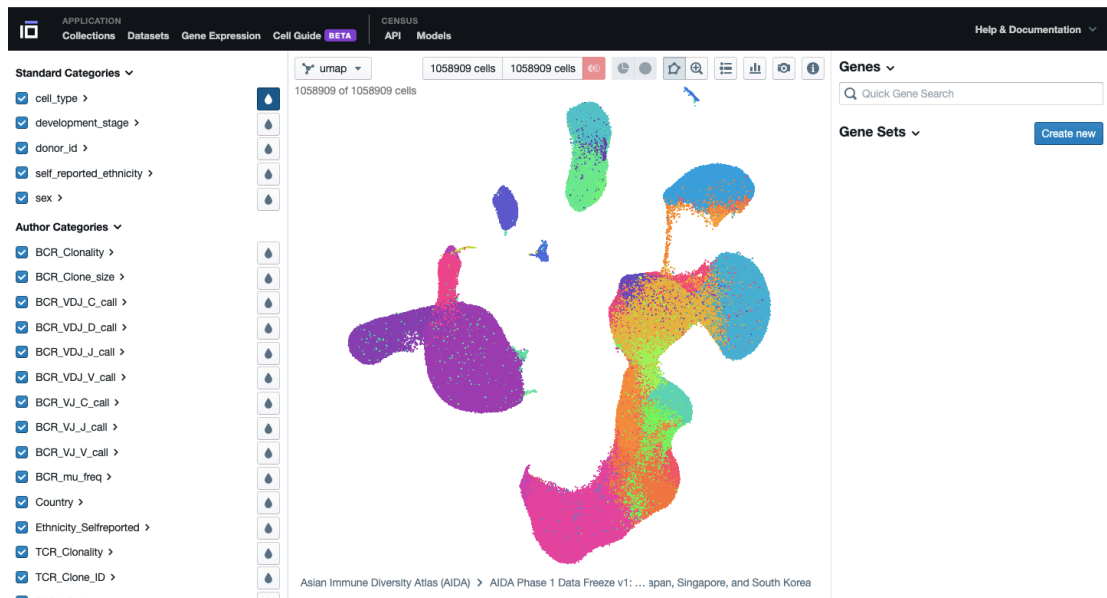
[Show Less](#)

Publication [Kock et al. \(2024\) bioRxiv](#)
Contact [Shyam Prabhakar](#)
Raw Data [explore.data.humancellatlas.org](#)

Dataset	Tissue	Disease	Assay	Organism	Cells	Download	Explore
AIDA Phase 1 Data Freeze v1: Chinese, Indian, Japanese, Korean, and Malay donors in Japan, Singapore, and South Korea	blood	normal	10x 5' v2	Homo sapiens	1,058,909	Download	Explore

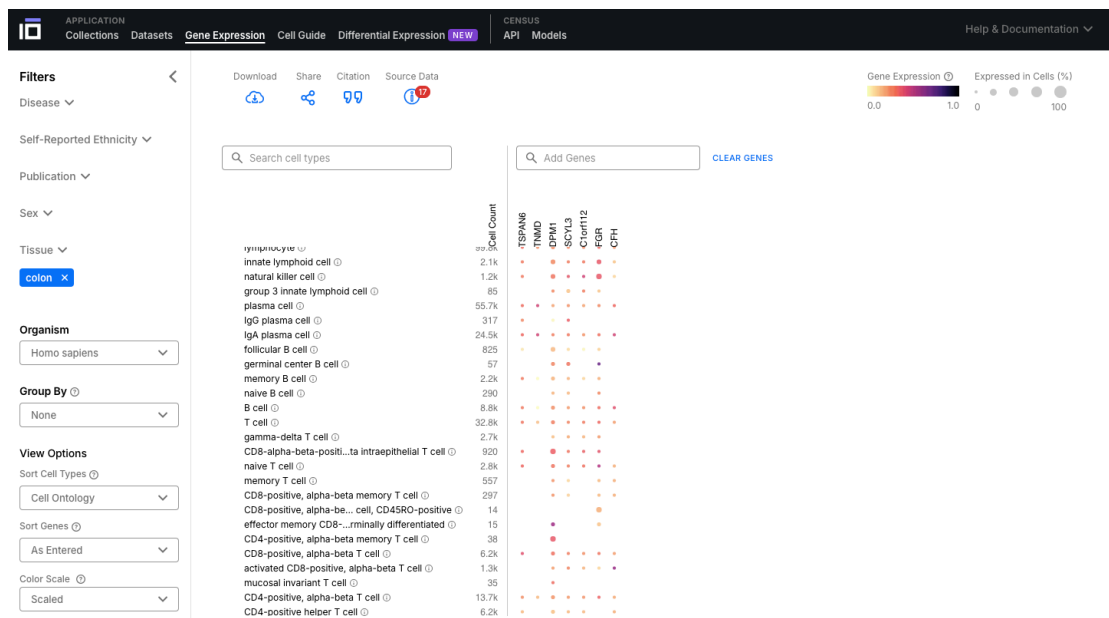
2. Explore a Dataset of Interest:

- Explore the different functionalities available. Navigate through the different categories, activating the collors for each cell available based on the cell type, developmental stage, ethnicity, sex, disease (if available), etc.
- Play with the cells, try to explore the different graphics available and to obtain different visualization of the data according to your preferences.
- Select two groups of cells of your interest to identify the top differentially expressed genes. Do it based on your preferences: cell type, sex, disease status, etc. Change the colors of one of the differentially expressed genes of interest.
- Play with the different functionalities available to further explore your dataset of interest.



3. Explore the Gene Expression Functionality:

- Select the dataset of interest based on the Filters; add a list of interest and visualize their expression across the different cell types.
- Explore the graphic, observing their expression levels and percentage of cells expressing that gene.



4. Explore the Cell Guide Functionality:

- Select a cell or a tissue of interest and navigate through its ontology.
- Cells can have different sub-types, so they are classified based on an ontology.
- The cell ontology (CL) is a standardized framework for describing and categorizing cell types based on their characteristics, functions, and relationships. It provides a common language and set of terms to define and annotate cell types across different species, tissues, and datasets.
- Navigate through the cell or tissue of your interest, exploring its description, ontology, marker genes and data available.

The screenshot shows the Cell Guide interface for the Fibroblast cell type. The page includes the following components:

- Header:**
 - APPLICATION: Collections, Datasets, Gene Expression, Cell Guide, Differential Expression, **NEW**
 - CENSUS: API, Models
 - Help & Documentation
- Cell Type:** Fibroblast (CL:0000057)
- Filters:**
 - Homo sapiens
 - Tissue Agnostic
- Search:**
- Description:**

Fibroblasts are a type of mesenchymal cell which constitute the most common cells of connective tissues in mammals. They maintain the structural integrity of connective tissues by continuously secreting precursors of the extracellular matrix. The matrix keeps tissues structurally sound, offers a scaffolding for cells and serves as a medium for the exchange of nutrients and waste between cells. Fibroblasts play a critical role in wound healing as well, where they migrate towards the site of injury and produce a new extracellular matrix.

Fibroblasts are resilient and versatile, revealing diverse variation in their shape and size, often reflecting their function and location in the body. They exhibit a spindle-like morphology, with a flat and elliptical nucleus located at the center. Characteristically, fibroblasts contain a large rough endoplasmic reticulum and abundant Golgi apparatus to facilitate protein synthesis for the extracellular matrix.

In addition to producing the extracellular matrix, fibroblasts possess the ability to differentiate into other cell types, showcasing their importance in tissue repair. For instance, when a tissue injury occurs, fibroblasts can transform into myofibroblasts, cells with features of both fibroblasts and smooth muscle cells, to aid in the wound healing process. They also play a role in immune response, as they can present antigens to T-cells and produce a variety of cytokines and chemokines.
- Validation:**
 - Validated** This description has been validated by our Biocurator team. If you believe a description is inaccurate, please [submit a correction](#).
- Synonyms:** N/A
- Citations:** [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#)
- Sidebar:**
 - Intro
 - Cell Ontology
 - Marker Genes
 - Data

Fibroblast

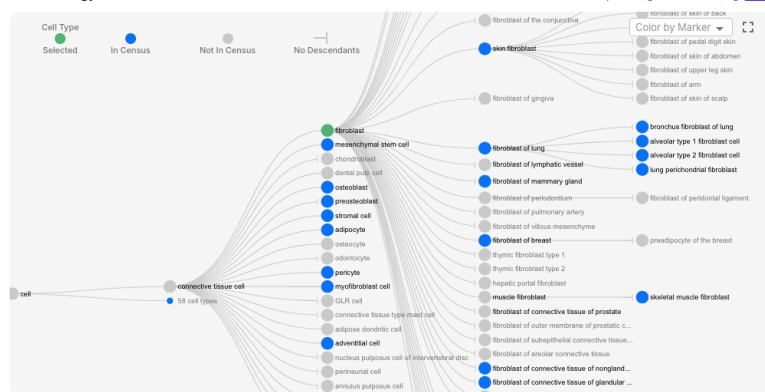
CL:0000057

Homo sapiens

Tissue Agnostic

Search cell types or tissues

Cell Ontology



Intro

Cell Ontology

Marker Genes

Data

Fibroblast

CL:0000057

Homo sapiens

Tissue Agnostic

Search cell types or tissues

Marker Genes

Symbol	Name	Effect Size	Specificity	Mean Expression	% of Cells
DCN	decorin	1.50	0.97	3.54	81.1
COL6A2	collagen type VI alpha 2 chain	1.12	0.98	2.48	69.4
LUM	lumican	1.01	0.96	2.92	57.1
COL1A2	collagen type I alpha 2 chain	1.00	0.96	3.00	65.9
COL6A3	collagen type VI alpha 3 chain	0.89	0.98	2.35	51.7
COL6A1	collagen type VI alpha 1 chain	0.83	0.98	2.28	55.0
COL3A1	collagen type III alpha 1 chain	0.78	0.95	3.08	53.9
C1S	complement C1s	0.77	0.96	2.32	47.0
FBLN1	fibulin 1	0.75	0.96	2.52	47.5
COL1A1	collagen type I alpha 1 chain	0.71	0.95	3.09	48.9

< 1 2 3 4 >

Source: CZI

Intro

Cell Ontology

Marker Genes

Data

Fibroblast

CL:0000057

Homo sapiens

Tissue Agnostic

Search cell types or tissues

Data

Collection	Publication	Tissue	Disease	Organism
A molecular cell atlas of the human lung from single cell RNA sequencing	Travaglini, Kyle J. et al. (2020) Nature	blood lung	normal	Homo sapiens
Construction of a human cell landscape at single-cell level	Han, Xiaoping et al. (2020) Nature	49 tissues	normal	Homo sapiens
Single cell transcriptional and chromatin accessibility profiling redefine cellular heterogeneity in the adult human kidney	Muto, Yoshiharu et al. (2021) Nat Commun	cortex of kidney	normal	Homo sapiens
Multimomics single-cell analysis of human pancreatic islets reveals novel cellular states in health and type 1 diabetes	Fasolino, Maria et al. (2022) Nat Metab	islet of Langerhans	normal type 1 diabetes mellitus	Homo sapiens
Cells of the adult human heart	Litviňuková, Monika et al. (2020) Nature	6 tissues	normal	Homo sapiens
Spatiotemporal analysis of human intestinal development at single-cell resolution	Fawcner-Corbett, David et al. (2021) Cell	colon hindgut ileum	normal	Homo sapiens
A human cell atlas of fetal gene expression	Cao, Junyue et al. (2020) Science	15 tissues	normal trisomy 18	Homo sapiens

Intro

Cell Ontology

Marker Genes

Data

5. Explore the Differential Expression Functionality:

- Select the groups you want to perform a differential expression analysis based on the organism, tissue, cell type, disease, ethnicity, sex, among other options available.

- In this example we identified the differentially expressed genes according to the sex in B cells from the colon.

This tool uses Welch's t-test to identify differentially expressed genes between groups of cells in the CELLxGENE Census. We recommend using this tool for preliminary investigations and following up with more robust methods for formal analysis. Learn more about data filtering and normalization [here](#).

Organism: Homo sapiens **Method:** T-test **Overlap Behavior:** [Buttons]

Cell Group 1: 4,155 cells
 Tissue: Search [colon] **Cell Type:** Search [B cell] Publications: Disease: Ethnicity: Sex: Search [female]

Cell Group 2: 4,668 cells
 Tissue: Search [colon] **Cell Type:** Search [B cell] Publications: Disease: Ethnicity: Sex: Search [male]

Find genes

Gene	Log Fold Change	Effect Size
e.g. JCHAIN	e.g. >1.0	e.g. >1.0
RN7SKP17...	0.732	0.673
XIST	0.299	0.564
IGHD	0.306	0.424
TCL1A	0.395	0.398
HLA-DQA1	0.447	0.371
NCL	0.340	0.326
HLA-DPA1	0.378	0.326
H1-2	0.186	0.316
HLA-DRB5	0.381	0.314
H1-4	0.171	0.309
HLA-DPB1	0.358	0.307
H3-3A	0.335	0.299
PTMA	0.250	0.294
CD79B	0.329	0.294

6. Explore the CELLxGENE Census:

- Get familiarized with the Census platform, which allow you to access, query, and analyze all single-cell RNA data from CELLxGENE.

cellxgene-census 1.15.0

Search docs

Installation
 Quick start
 What's new?
 Census in AWS
 Census data and schema
 Census data releases
 Python API
 Python tutorials
 R API & tutorials
 FAQ
 Release notes

CZ CELLxGENE Discover Census

New to the Census: Train PyTorch models directly with Census data with our efficient and easy-to-use PyTorch loaders. [Learn more!](#)

Explore benchmarks of Census models and embeddings. [See the report!](#)

CZ CELLxGENE Discover Census

The Census provides efficient computational tooling to **access, query, and analyze all single-cell RNA data from CZ CELLxGENE Discover**. Using a new access paradigm of cell-based slicing and querying, you can interact with the data through TileDB-SOMA, or get slices in AnnData, Seurat, or SingleCellExperiment objects, thus accelerating your research by significantly minimizing data harmonization.

Get started:

- [Quick start \(Python and R\)](#)
- [Census data and schema](#)
- [Python tutorials](#)
- [R tutorials](#)
- [Github repository](#)

CZ CELLxGENE DISCOVER

Single-cell RNA datasets

Dataset 1

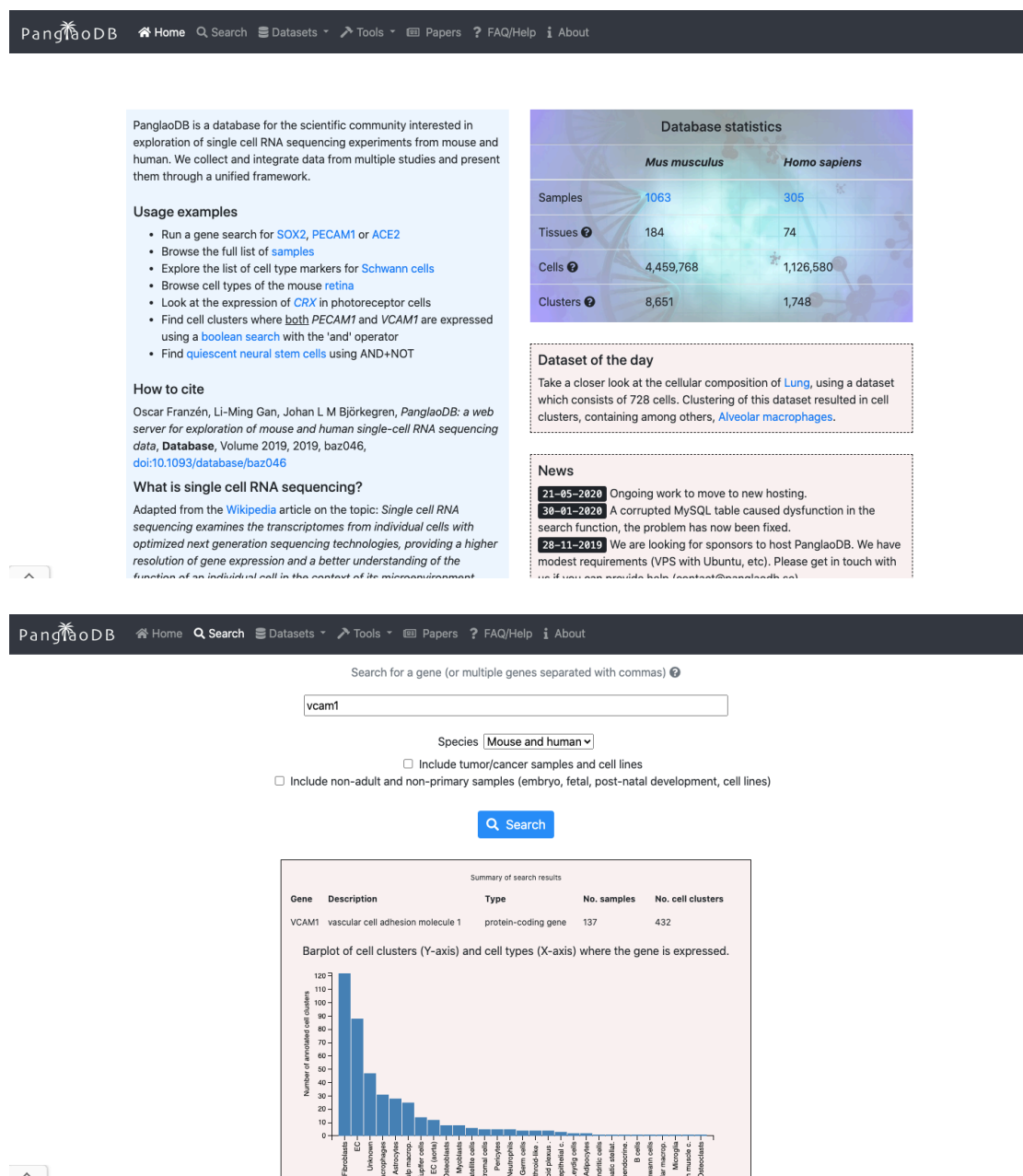
Census (powered by TileDB)

>60K Genes

7. Now enjoy the repository, by playing with the data and genes of interest.

Panglao DB

Brief Description:



Cell type gene expression markers

This is a list of gene expression markers are used to define cell types. Green rows indicate canonical markers (classical markers used to define the cell type).

Summary: 8286 associations (178 cell types, 4679 gene symbols, 29 tissues); Last updated: 27/03/2020 10:44:00 CET

Filter

Show cell type:

[get tsv file](#) [add marker](#)

Gene expression markers for Erythroblasts

Vote(s)	Species	Official gene symbol	UI	Sensitivity (human)	Sensitivity (mouse)	Specificity (human)	Specificity (mouse)	Marker count	Cell type	Germ layer	Organ	Aliases	Product description	Disease	Action
2	Mm Hs	TFRC	0.042	NA	NA	0.061	NA	4	Erythroblasts	Mesoderm	Blood	CD71, TFR1, p90	transferrin receptor	Y	flag
2	Mm Hs	CGA	0.003	NA	NA	0.008	0.001	2	Erythroblasts	Mesoderm	Blood	GPHA, GPHA1, F9, HALHA, TSHA	glycoprotein hormones, alpha polypeptide		flag
2	Mm Hs	GYP A	0.011	NA	NA	0.027	0.008	2	Erythroblasts	Mesoderm	Blood	GPA, CD235a	glycophorin A (MNS blood group)	Y	flag
1	Mm Hs	BCL11A	0.134	NA	NA	0.113	0.143	4	Erythroblasts	Mesoderm	Blood	BCL11A-XL, BCL11A-L, BCL11A-S, CTIP1, HBFQTL11A, S, ZNF956, EVI9	B cell CLL/lymphoma 11A	Y	flag
1	Mm Hs	BPGM	0.063	NA	NA	0.046	0.069	2	Erythroblasts	Mesoderm	Blood		bisphosphoglycerate mutase	Y	flag
1	Mm Hs	KIT	0.015	NA	NA	0.003	0.019	10	Erythroblasts	Mesoderm	Blood	CD117, SCFR, KIT, KIT	pro-oncogene receptor tyrosine kinase	Y	flag
1	Mm Hs	IGBP1	0.089	NA	NA	0.129	0.071	1	Erythroblasts	Mesoderm	Blood		immunoglobulin binding protein 1	Y	flag

2. Now enjoy the repository, by playing with the data and genes of interest.

CellTypist

Brief Description:

CellTypist is a web-based platform designed to facilitate cell type identification, classification, and annotation. It provides a user-friendly interface for researchers to annotate and classify cell types on their own data.

Practical Exercises:

1. Explore the General Portal Interface:

- Go to the CellTypist website: <https://www.celxltypist.org/>
- Navigate through the available Encyclopedia. Explore deeper a group of cells of your interest.
- In "Resources", navigate through the models available; explore the organs there available, as well as have access to the Python package.
- At the "Home" web page, explore the automatic tool to annotate your own data.
- Access the available tutorials to dive into the platform functionalities.

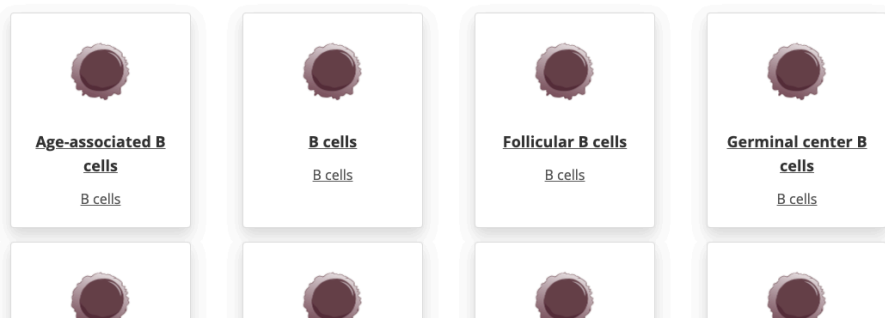
Automated cell type annotation for scRNA-seq datasets

[Annotate your dataset NOW](#)
[Tutorials](#)
[Cell type encyclopedia](#)
[Cell/Gene search](#)


Cell types

Genes

All information below comes from our encyclopedia table which can be downloaded [here](#).


[Provide feedback on this cell type](#)

High hierarchy

B cells

Description

B lymphocytes with diverse cell surface immunoglobulin receptors recognising specific antigenic epitopes and mediating humoral immunity

Low hierarchy

B cells

Curated Markers

[CD79A](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

[MS4A1](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

[CD19](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

Top Model Markers

[HBG2](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

[HLA-DQA2](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

[HBA2](#)

[GeneCards](#) [NCBI](#) [Ensembl](#)

Models

How are the models trained?

All models are built on the logistic regression framework. Traditional logistic regression will be used in most cases. SGD learning can be optionally implemented depending on the size of the training dataset. For example, when the training dataset contains a huge number of cells, the data can be modelled with SGD logistic regression using mini-batch training. Briefly, in each epoch cells are shuffled and binned into equal-sized mini-batches (1,000 cells per batch), and later are sequentially trained by 100 such batches randomly sampled out of all batches. This process is repeated for 10–30 epochs. In addition to the models listed below, users can train their own [custom models](#).

Which model should I select?

Models are usually selected based on the context of the query data. For example, for immune cell types it is recommended to start from the "Immune_All_Low/High" models as they contain immune cell types collected from different tissues. The "Low" indicates low-hierarchy (high-resolution) cell types and subtypes, and the "High" indicates high-hierarchy (low-resolution) ones.

Organ atlas

CellHint auto-harmonised and auto-integrated atlas

Blood	Bone marrow	Heart	Hippocampus
			
UMAP (CELLxGENE)	UMAP (CELLxGENE)	UMAP (CELLxGENE)	UMAP (CELLxGENE)
Tree plot	Tree plot	Tree plot	Tree plot
Download data	Download data	Download data	Download data
Download models	Download models	Download models	Download models

Get Started

[Online](#) [Python](#) [_ Command-line](#)

Try it online

Email Address

Select Model [?]

Majority Voting [?]

We'll send the results to your email.
Detail of each model can be found [here](#).
Check how majority voting works [here](#).

Only .csv and .h5ad file types accepted for online analysis. Desired format can be found [here](#).

2. Now enjoy the platform, by playing with the data and genes of interest.

GEO

Brief Description:

GEO is a comprehensive public database that archives and freely distributes microarray, next-generation sequencing, and other forms of high-throughput functional genomic data. It is an invaluable resource for researchers, supporting discovery of new insights into gene function, regulation and expression; supporting the data reuse.

Practical Exercises:

1. Explore the General Portal Interface:

- Go to the GEO website: <https://www.ncbi.nlm.nih.gov/geo/>
- Search for a dataset of interest, using for example the following keywords: "single cell heart"
- Explore the public datasets related to heart single-cell experiments (or related to the keywords used). Select one to obtain further information from that study.
- Navigate through the samples available.
- Evaluate if they followed the FAIR principles when depositing their data.

The screenshot shows the Gene Expression Omnibus (GEO) website. At the top, there's a navigation bar with links for NCBI, Resources, and How To. Below this is a search bar with the text "single-cell heart" and a "Search" button. The main content area is titled "Gene Expression Omnibus" and includes a brief description of the database. Below the description, there are three main sections: "Getting Started", "Tools", and "Browse Content". The "Getting Started" section includes links for Overview, FAQ, About GEO DataSets, About GEO Profiles, About GEO2R Analysis, How to Construct a Query, and How to Download Data. The "Tools" section includes links for Search for Studies at GEO DataSets, Search for Gene Expression at GEO Profiles, Search GEO Documentation, Analyze a Study with GEO2R, Studies with Genome Data Viewer Tracks, Programmatic Access, FTP Site, and ENCODE Data Listings and Tracks. The "Browse Content" section includes a Repository Browser and a table showing the number of DataSets (4348), Series (233869), Platforms (26423), and Samples (7346330). At the bottom, there's a section for "Information for Submitters" with links for Login to Submit, Submission Guidelines, Update Guidelines, MIAME Standards, Citing and Linking to GEO, Guidelines for Reviewers, and GEO Publications. The footer contains links for NLM, NIH, Email GEO, Disclaimer, Accessibility, and HHS Vulnerability Disclosure.

NCBI Resources How To Sign in to NCBI

GEO Home Documentation Query & Browse Email GEO

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.

single-cell heart Search

Getting Started

- Overview
- FAQ
- About GEO DataSets
- About GEO Profiles
- About GEO2R Analysis
- How to Construct a Query
- How to Download Data

Tools

- Search for Studies at GEO DataSets
- Search for Gene Expression at GEO Profiles
- Search GEO Documentation
- Analyze a Study with GEO2R
- Studies with Genome Data Viewer Tracks
- Programmatic Access
- FTP Site
- ENCODE Data Listings and Tracks

Browse Content

Repository Browser

DataSets:	4348
Series:	233869
Platforms:	26423
Samples:	7346330

Information for Submitters

Login to Submit	Submission Guidelines	MIAME Standards
	Update Guidelines	Citing and Linking to GEO
		Guidelines for Reviewers
		GEO Publications

NLM | NIH | Email GEO | Disclaimer | Accessibility | HHS Vulnerability Disclosure |

GEO DataSets

GEO DataSets

Search

Create alert Advanced

Help

Entry type
DataSets (32)
Series (1,678)
Samples (17,586)
Platforms (25)

Organism
Customize ...

Study type
Expression profiling by array
Methylation profiling by array
Customize ...

Author
Customize ...

Attribute name
tissue (14,988)
strain (7,730)
Customize ...

Publication dates
30 days
1 year
Custom range...

[Clear all](#)

[Show additional filters](#)

Summary 20 per page Sort by Default order

Send to:

Filters: [Manage Filters](#)

Search results

Items: 1 to 20 of 19321

<< First < Prev Page 1 of 967 Next > Last >>

☐ [Spatially clustered type I interferon responses at injury borderzones](#)

1. (Submitter supplied) Sterile inflammation after myocardial infarction (MI) is classically credited to myeloid cells interacting with dead cell debris in the infarct zone (IZ). Here, we show that borderzone cardiomyocytes are the dominant initiators of a novel type I interferon (IFN) response in the infarct borderzone zone (BZ). Using spatial transcriptomics analysis of mice and humans, we find that MI induces colonies of interferon induced cells (IFNICs) expressing interferon stimulated genes (ISGs) decorating the BZ, where cardiomyocytes experience mechanical stress, nuclear rupture, and escape of chromosomal DNA. [more...](#)

Organism: Homo sapiens; Mus musculus
Type: Expression profiling by high throughput sequencing
Platforms: GPL34290 GPL24247 GPL24676 37 Samples

[Download data](#)

Series Accession: GSE269054 ID: 200269054

☐ [Spatially clustered type I interferon responses at injury borderzones \[scRNA-Seq\]](#)

2. (Submitter supplied) Sterile inflammation post-myocardial infarction was thought to be driven by myeloid cells, however, we show that borderzone cardiomyocytes are the critical initiators of the Type I Interferon response through spatial, single-cell, and single-nuclei transcriptomics.

Organism: Mus musculus
Type: Expression profiling by high throughput sequencing
Platform: GPL24247 12 Samples

[Download data](#)

Series Accession: GSE268876 ID: 200268876

☐ [Blood-Generating Heart-Forming Organoids \(BG-HFOs\) recapitulate co-development of the hematopoietic system and the embryonic heart](#)

3. (Submitter supplied) Hematopoietic stem cells (HSCs) derive from the hemogenic endothelium of the dorsal aorta, which develops in spatial and functional proximity to the heart anlagen in the human embryo. Despite the biomedical importance of HSCs, their in vitro induction and niche-like stabilization in a proper developmental context is challenging and poorly established. The underlying reasons include the poorly understood signals controlling HSC formation and the lack of unequivocal markers specifying their stages of differentiation and activation. [more...](#)

Top Organisms [\[Tree\]](#)

Mus musculus (9939)
Homo sapiens (3495)
Pleurodeles walli (3068)
Rattus norvegicus (777)
Ambystoma mexicanum (613)
[More...](#)

Find related data

Database:

[Find items](#)

Search details

```
("single person"[MeSH Terms] OR
single[All Fields]) AND ("cells"
[MeSH Terms] OR cell[All Fields])
AND ("heart"[MeSH Terms] OR
heart[All Fields])
```

[Search](#)

[See more...](#)

Important Links

[GEO Home](#)

[GEO Documentation](#)

[About GEO DataSets](#)

[Construct a Query](#)

[Download Options](#)

GEO help: Mouse over screen elements for information.

Scope: Format: Amount: GEO accession:

Series GSE268876
[Query DataSets for GSE268876](#)

Status	Public on Aug 06, 2024
Title	Spatially clustered type I interferon responses at injury borderzones [scRNA-Seq]
Organism	Mus musculus
Experiment type	Expression profiling by high throughput sequencing
Summary	Sterile inflammation post-myocardial infarction was thought to be driven by myeloid cells, however, we show that borderzone cardiomyocytes are the critical initiators of the Type I Interferon response through spatial, single-cell, and single-nuclei transcriptomics.
Overall design	Nuclei and cells were isolated from 12-14-week-old adult wild-type C57BL/6J, IFNAR ^{-/-} , and IRF3 ^{-/-} mice. For whole cell suspensions, freshly harvested hearts were enzymatically digested prior to filtering and flow cytometry sorting. Nuclei were isolated from fresh frozen tissue, minced and lysed, then filtered prior to collection.
Contributor(s)	Ninh VK , Calcagno DM , Yu JD , Zhang B , Taghdiri N , Sehgal R , Mesfin JM , Chen CJ , Kalhor K , Toomu A , Duran JM , Adler E , Hu J , Zhang K , Christman KL , Fu Z , Bintu B , King KR
Citation missing	<i>Has this study been published? Please login to update or notify GEO.</i>
Submission date	Jun 03, 2024
Last update date	Aug 06, 2024
Contact name	Kevin Robert King
E-mail(s)	krking@health.ucsd.edu
Organization name	UCSD
Department	Medicine
Lab	King
Street address	9500 Gilman Dr.
City	La Jolla
State/province	CA
ZIP/Postal code	92161
Country	USA

Platforms (1) [GPL24247](#) Illumina NovaSeq 6000 (Mus musculus)

Samples (12) [GSM8301921](#) IfnarKO_D4_WC1, biol rep1
[Less...](#) [GSM8301922](#) IfnarKO_D4_WC2, biol rep2

[GSM8301923](#) WT_D4_WC1, biol rep1

[GSM8301924](#) WT_D4_WC2, biol rep2

[GSM8301925](#) WT_D4_WC3, biol rep3




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NCBI > GEO > **Accession Display**
Not logged in | [Login](#)

GEO help: Mouse over screen elements for information.

Scope:
Format:
Amount:
GEO accession:

Sample GSM8301921
[Query DataSets for GSM8301921](#)

Status	Public on Aug 06, 2024
Title	IfnarKO_D4_WC1, biol rep1
Sample type	SRA
Source name	Left Ventricle
Organism	Mus musculus
Characteristics	tissue: Left Ventricle strain: C57BL/6J genotype: IFNAR -/- treatment: left anterior descending artery permanent ligation
Extracted molecule	cytoplasmic RNA
Extraction protocol	For whole cell suspensions, freshly harvested hearts were enzymatically digested prior to filtering and flow cytometry sorting. Nuclei were isolated from fresh frozen tissue, minced and lysed, then filtered prior to collection. Single cell and nuclei libraries were prepared according to standard 10X Genomics protocols Dual index Illumina single cell RNA sequencing
Library strategy	RNA-Seq
Library source	transcriptomic single cell
Library selection	cDNA
Instrument model	Illumina NovaSeq 6000
Description	10X Genomics IfnarKO_D4_WC1, biol rep1
Data processing	The alignments, filtering, and demultiplexing of generated fastq files were run with the 10X Genomics Cell Ranger analysis pipeline using standard configurations (https://support.10xgenomics.com/single-cell-gene-expression/software/pipelines/latest/what-is-cell-ranger). Variations in sequencing depth were accounted for by normalization with Seurat's SCTransform v2, based on negative binomial models which utilize a lower bound on the gene-level standard deviations. Multiple single-cell RNA-seq datasets were integrated in Seurat to allow harmonized clustering and downstream analysis across conditions. Anchoring cell pairs between the datasets were identified with Canonical Correlation Analysis (CCA) and Mutual Nearest Neighbors (MNN) methods with the Seurat FindIntegrationAnchors function. After scaling and centering expression values for each variable gene, the integrated data underwent linear dimensionality reduction using principal component analysis (PCA). Clustering was performed through the Shared Nearest Neighbor (SNN) clustering algorithm with Louvain method for modularity optimization using the Seurat FindNeighbors and FindClusters functions. Two-dimensional data visualization was done by performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction. and

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SRA

Brief Description:

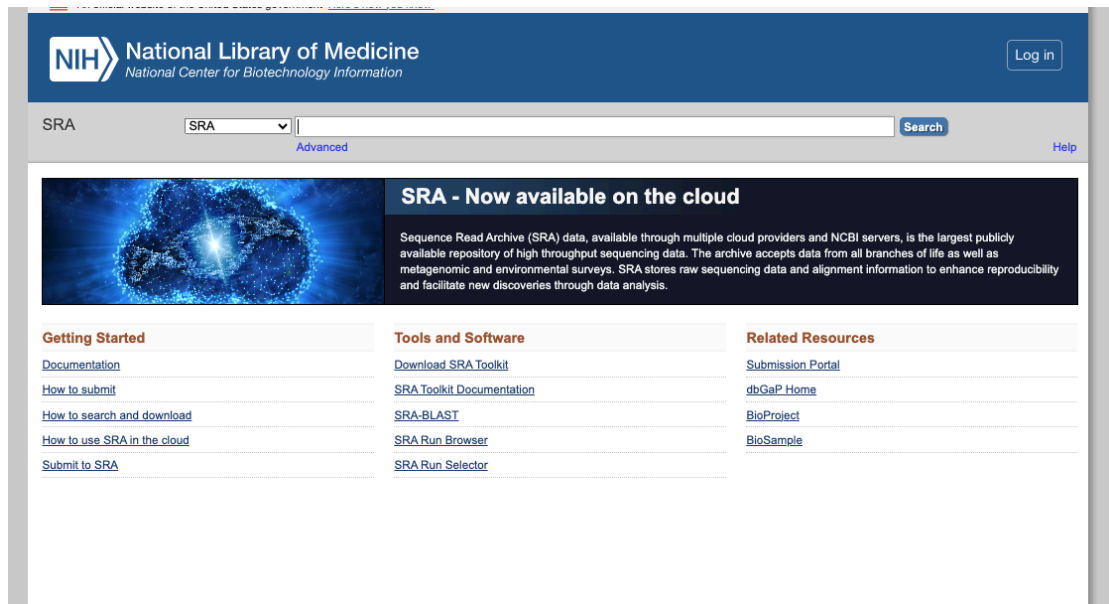
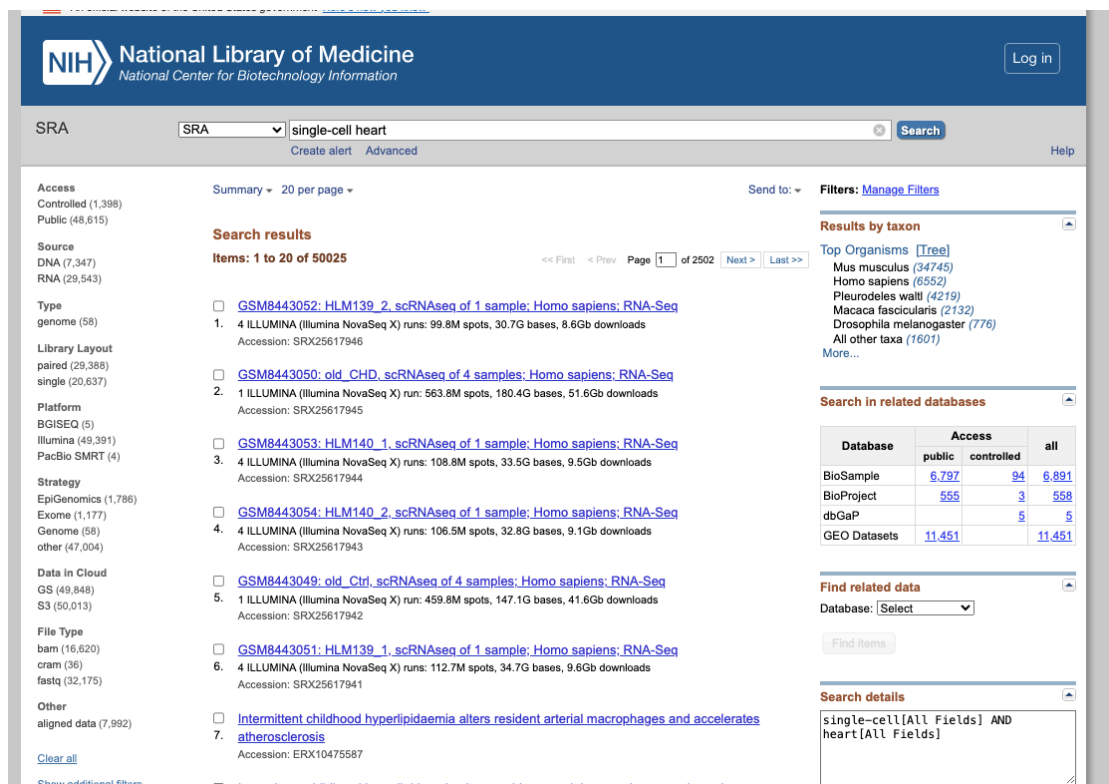
SRA is a comprehensive public database that archives and freely distributes high-throughput sequencing data, including RNA-seq, DNA-seq, and other forms of next-generation sequencing (NGS) data.

Practical Exercises:

1. Explore the General Portal Interface:

- Go to the SRA website: <https://www.ncbi.nlm.nih.gov/sra>

- Search for a dataset of interest, using for example the following keywords: "single cell heart"
- Filter the result according to the source, sequencing platform, organism of interest, or other features available.
- Explore the public datasets related to heart single-cell experiments (or related to the keywords used). Select one to obtain further information from that study.
- Navigate through the samples available.
- Evaluate if they followed the FAIR principles when depositing their data.

Access
Controlled (1,398)
Public (48,615)

Source
DNA (7,347)
RNA (29,543)

Type
genome (58)

Library Layout
paired (29,388)
single (20,637)

Platform
BGISEQ (5)
Illumina (49,391)
PacBio SMRT (4)

Strategy
EpiGenomics (1,786)
Exome (1,177)
Genome (58)
other (47,004)

Data in Cloud
GS (49,849)
S3 (50,013)

File Type
bam (16,620)
cram (36)
fastq (32,175)

Other
aligned data (7,992)

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Search results
Items: 1 to 20 of 50025

1. [GSM8443052: HLM139_2_scRNAseq of 1 sample: Homo sapiens: RNA-Seq](#)
4 ILLUMINA (Illumina NovaSeq X) runs: 99.8M spots, 30.7G bases, 8.6Gb downloads
Accession: SRX25617946

2. [GSM8443050: old_CHD_scRNAseq of 4 samples: Homo sapiens: RNA-Seq](#)
1 ILLUMINA (Illumina NovaSeq X) run: 563.8M spots, 180.4G bases, 51.6Gb downloads
Accession: SRX25617945

3. [GSM8443053: HLM140_1_scRNAseq of 1 sample: Homo sapiens: RNA-Seq](#)
4 ILLUMINA (Illumina NovaSeq X) runs: 108.8M spots, 33.5G bases, 9.5Gb downloads
Accession: SRX25617944

4. [GSM8443054: HLM140_2_scRNAseq of 1 sample: Homo sapiens: RNA-Seq](#)
4 ILLUMINA (Illumina NovaSeq X) runs: 106.5M spots, 32.8G bases, 9.1Gb downloads
Accession: SRX25617943

5. [GSM8443049: old_Ctrl_scRNAseq of 4 samples: Homo sapiens: RNA-Seq](#)
1 ILLUMINA (Illumina NovaSeq X) run: 459.8M spots, 147.1G bases, 41.6Gb downloads
Accession: SRX25617942

6. [GSM8443051: HLM139_1_scRNAseq of 1 sample: Homo sapiens: RNA-Seq](#)
4 ILLUMINA (Illumina NovaSeq X) runs: 112.7M spots, 34.7G bases, 9.6Gb downloads
Accession: SRX25617941

7. [Intermittent childhood hyperlipidaemia alters resident arterial macrophages and accelerates atherosclerosis](#)
Accession: ERX10475587

8. [Intermittent childhood hvoerlipidaemia alters resident arterial macrophages and accelerates](#)

Results by taxon
Top Organisms [Tree](#)
Mus musculus (34745)
Homo sapiens (6552)
Pleurodeles waltl (4219)
Macaca fascicularis (2132)
Drosophila melanogaster (776)
All other taxa (1601)
[More...](#)

Search in related databases

Database	Access		all
	public	controlled	
BioSample	6,797	94	6,891
BioProject	555	3	558
dbGaP		5	5
GEO Datasets	11,451		11,451

Find related data
Database: [Select](#)

[Find items](#)

Search details
single-cell[ALL Fields] AND heart[ALL Fields]



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SRX25617946: GSM8443052: HLM139_2, scRNAseq of 1 sample; Homo sapiens; RNA-Seq
4 ILLUMINA (Illumina NovaSeq X) runs: 99.8M spots, 30.7G bases, 8.6Gb downloads

External Id: GSM8443052_r1

Submitted by: Hannover Medical School

Study: Early-life thymectomy leads to an increase of granzyme-producing ?d T cells in children with congenital heart disease
[PRJNA1144917](#) • [SRP524526](#) • [All experiments](#) • [All runs](#)
[show Abstract](#)

Sample: HLM139_2, scRNAseq of 1 sample
[SAMN43048256](#) • [SRS22263297](#) • [All experiments](#) • [All runs](#)
[Organism:](#) [Homo sapiens](#)

Library:
Name: GSM8443052
Instrument: Illumina NovaSeq X
Strategy: RNA-Seq
Source: TRANSCRIPTOMIC SINGLE CELL
Selection: cDNA
Layout: PAIRED
Construction protocol: The PBMCs were thawed and fluorescence-activated cell sorting (FACS) was performed to isolate live CD3+TCRβ-TCRγδ+ cells and live CD3+TCRαβ+TCRγδ- cells. These sorted T cells were then used to generate single-cell RNA sequencing (scRNAseq) libraries with the Chromium Next GEM Single Cell V(D)J Reagent Kits v1.1 (for infants) and v2 (for 5-11-year-old children), following the manufacturer's protocol (10x Genomics).

Runs: 4 runs, 99.8M spots, 30.7G bases, 8.6Gb

Run	# of Spots	# of Bases	Size	Published
SRR30150116	23,485,897	7.2G	2Gb	2024-08-07
SRR30150119	20,621,771	6.4G	1.8Gb	2024-08-07
SRR30150120	24,128,921	7.4G	2.1Gb	2024-08-07
SRR30150121	31,588,887	9.7G	2.7Gb	2024-08-07

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[BioSample](#)
[GEO DataSets](#)
[Taxonomy](#)

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SRA

IfnarKO_D4_WC1, biol rep1

biosample

single cell heart (19321)

GEO DataSets

SRA570744 (1)

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Figure 3.1, [Morphology of normal colon tissue...]. - Dynamics of Cancer

See more...

ID: 34625204

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NOTE

In addition, there is **SRA Explorer**, an interactive SRA data visualization tool, which facilitates navigation and access to raw data stored in the SRA, allowing for efficient data searching and downloading.